

Project Requirements and Dependencies

1. Python Environment This project was developed and tested using **Python 3.10** && **VS Code** (Python 3.8 or higher is recommended).

2. Required Libraries To successfully run the scripts and reproduce the results, the following third-party Python packages must be installed:

1. pandas (for data manipulation and reading CSV files)
2. numpy (for numerical operations and calculating averages/RMSE)
3. scikit-learn (for instantiating and training the Linear Regression, Decision Tree, and Random Forest models)
4. scipy (for performing the Wilcoxon Signed-Rank statistical test)
5. matplotlib (for generating the comparison bar charts)

3. Installation Command You can install all the required dependencies at once using the pip package manager. **Run the following command** in your terminal or command prompt:

```
pip install pandas==2.3.3 numpy==2.26 scikit-learn==1.7.2 scipy==1.15.3  
matplotlib==3.10.8
```